

KARAN RAJPUT

📍 Rajsamand, India 📞 +91-8824806030 ✉️ rajputkaran997@gmail.com
🌐 leetcode in linkedin 🐙 github

Profile Summary

Integrated M.Tech student in Computational Data Science with strong foundations in Machine Learning, Natural Language Processing, Data Structures Algorithms, and Software Engineering. Experienced in building end-to-end ML pipelines, NLP-based text classification systems, and LLM-powered applications using Python, Scikit-learn, HuggingFace, and modern AI workflows. Skilled in data preprocessing, feature engineering, model evaluation, experimentation, and AI solution development. Passionate about applying machine learning and generative AI to solve real-world problems at scale.

Education

- **Vellore Institute of Technology** (Oct 2022 – 2027)
Integrated M.Tech in Computer Science and Engineering *CGPA: 8.62/10*

Technical Skills

- **Programming:** Java, Python, SQL
- **Data Science:** Pandas, NumPy, Data Analysis, Data Preprocessing, Statistical Analysis
- **Software Engineering:** REST APIs, Spring Boot (Basics), Backend Development, ETL Workflows, Data Pipelines, Modular Programming, Version Control
- **Tools Platforms:** Git, GitHub, Linux, Shell Scripting (Basics), Jupyter Notebook, Google Colab, VS Code, Docker (Basics), Cloud Fundamentals (AWS)
- **Machine Learning AI:** Scikit-learn, NLP, TF-IDF, Text Classification, Feature Engineering, Model Evaluation, LLMs, RAG, Prompt Engineering, HuggingFace, Transformers

Projects

AI-Driven News Credibility and Influence Analysis *Jan 2026 – Present*

Python, NLP, Scikit-learn, HuggingFace, Pandas

- Built an NLP pipeline to classify credible and non-credible news articles using machine learning techniques.
- Processed and analyzed 45,000+ news records through automated preprocessing and feature engineering workflows.
- Developed TF-IDF feature extraction pipelines and trained Logistic Regression and SVM models.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-Score metrics while performing comparative error analysis.
- Utilized HuggingFace datasets and reproducible experimentation workflows for scalable model development.

AI-Powered Test Plan Generator and Debugging Assistant *May 2026 – Present*

Python, LLMs, RAG, Prompt Engineering, Streamlit, OpenAI API

- Developing an AI assistant that generates structured software test plans from project requirements and documentation.
- Implemented Retrieval-Augmented Generation (RAG) workflows to provide context-aware responses using uploaded documents.
- Built PDF ingestion, preprocessing, and semantic retrieval pipelines for knowledge extraction and search.
- Developed debugging assistance features capable of analyzing code snippets and suggesting corrective actions.
- Designed an interactive Streamlit interface for document upload, retrieval, and AI-assisted testing workflows.

Student Performance Analysis *Oct 2025*

Python, Pandas, NumPy, Matplotlib

- Performed exploratory data analysis on academic datasets to identify performance trends and patterns.
- Cleaned, transformed, and analyzed structured datasets using statistical techniques and data preprocessing workflows.
- Created visualizations and analytical reports to communicate insights and support data-driven decision making.

Certifications

- Introduction to Machine Learning – NPTEL (IIT Madras)
- Python for Data Science – Coursera

Languages

- English, Hindi